

A turning point for Alzheimer's disease?



Despite an archive of over 73,000 research papers published in the last two decades on the subject of Alzheimer's disease (AD), little clinical progress has been made relative to how people get sporadic AD and what can be done to help them avoid it. This review spotlights strategic steps that could be a turning point in the dramatic lowering of Alzheimer prevalence. The main strategy includes application of four pillars of prevention: 1) early identification of AD vascular risk factors; 2) early detection of AD vascular risk factors; 3) early intervention of AD vascular risk factors based on evidence-based medical decisions; 4) patient follow-up to assess and modify interventions as needed. Tandem to these four pillars of prevention, a proactive lifestyle consisting of a healthy diet coupled to physical and mental activity should be applied as part of any therapeutic intervention. We are persuaded by mounting and compelling evidence that AD is a multifactorial disorder kindled by vascular risk factors that generate chronic brain hypoperfusion (CBH) during advanced aging. A pathobiological cascade of biochemical events in the presence of CBH that leads to oxidative stress and neurodegeneration appears to involve multiple biofactors including micronutrients, trace metals, lipids, and pro-oxidants, as reviewed in this special issue of BioFactors. Modulation of these biofactors may help prevent or control incipient AD. © 2012 International Union of Biochemistry and Molecular Biology, Inc. Copyright © 2012 International Union of Biochemistry and Molecular Biology, Inc.